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Novel Control and Recovery Drilling Measures: the Key to Safely Drill the Structurally Complex and Depleted Overpressured-reservoirs in the South of Oman

M. Senani, M. Thuhli, Y. Abri, S. Farsi, Y. Abri, H. Rubaiey, M. Omairi, P. Lambkin, Q. Riyami, H. Bulushi, A. Mahrooqi, R. Siyabi, H. Balushi, V. S. Saroj, and A. Ponomarev, Petroleum Development Oman, Muscat, Sultanate of Oman

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Abstract

Salt-encased carbonate reservoirs in the South Oman Salt Basin (SOSB) present unique drilling hazards due to their potential for pressure compartmentalization. This complexity is amplified in mature fields where decades of production have caused significant depletion in most producing zones, while some compartments retain near-original pressures. Such pressure contrasts can narrow or eliminate safe drilling windows, increasing the risk of kicks, losses, or both.

In October 2023, the WELL-A well encountered original reservoir pressure of 19.2 kPa/m in a target compartment predicted to be in the range of 8 to 14 kPa/m. The unexpected high-pressure zone triggered a well control incident, highlighting limitations in the prevailing pore pressure prediction (PPP) workflow and the need for a structured approach to compartmentalization risk management.

A dedicated project was initiated to embed the lessons from WELL-A into the well planning and execution process. This involved developing a Compartmentalization Likelihood Tool, refining the PPP methodology to account for geological isolation scenarios, and enhancing drilling contingency protocols. The workflow integrates geological and structural boundary analysis, sealing potential evaluation, and normalization for field history to produce a qualitative compartmentalization likelihood classification—Very Unlikely, Unlikely, or Likely—that guides PPP design and drilling strategy.

The revised approach was piloted across multiple wells, successfully navigating wide pore pressure ranges by combining tailored mud weight envelopes, staged penetration strategies, and predefined safe exit procedures. In each case, the methodology enabled early recognition of high-pressure risks, better alignment of PPP with geological reality, and improved operational readiness.

This paper details the geological context, methodology, case study results, and practical lessons learned, providing a replicable framework for managing pressure uncertainty in depleted, structurally complex carbonate reservoirs. The approach is transferable to other mature salt-bounded systems globally, offering a pathway to safer and more predictable drilling operations in environments with elevated compartmentalization risk.

Introduction

Salt-encased carbonate reservoirs are among the most challenging hydrocarbon systems to drill and develop due to a combination of geological, geomechanical, and operational factors. Within the South Oman Salt Basin (SOSB) (Figure. 1), the Ara stringer reservoirs represent discrete carbonate slabs encapsulated by thick layers of salt (Figure. 2). This unique setting has enabled these reservoirs to preserve hydrocarbons for millions of years, but it has also created the possibility of isolated pressure compartments separated by sealing boundaries.

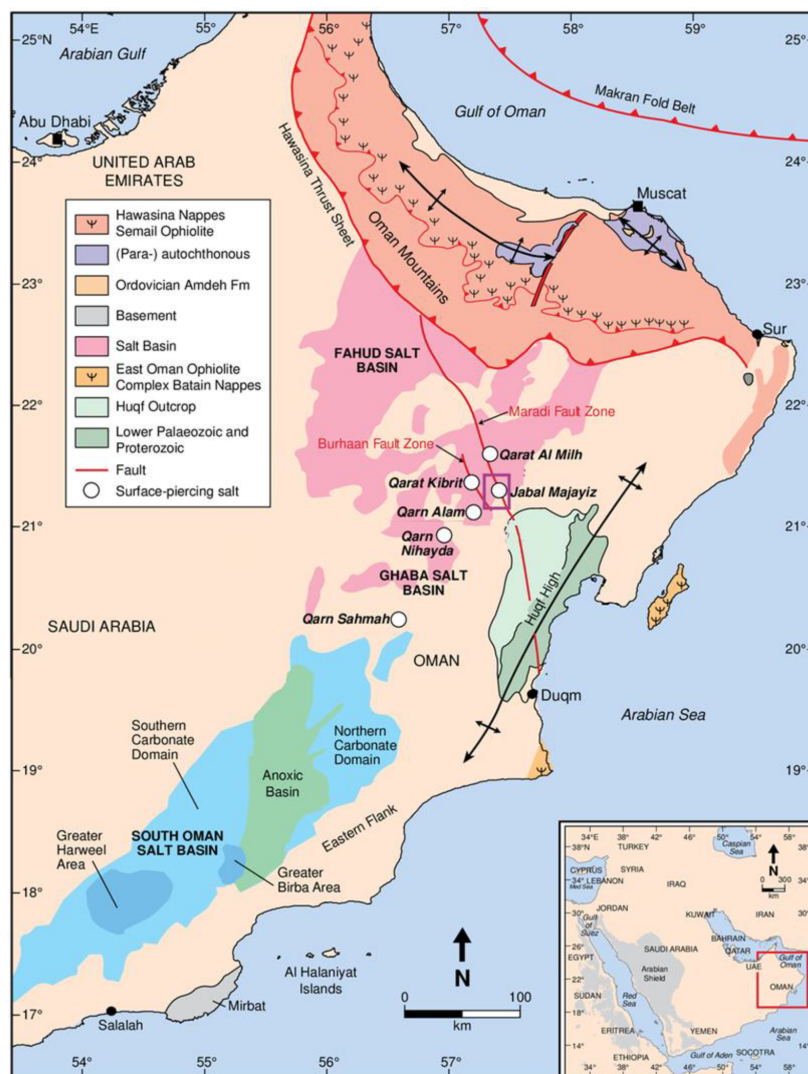


Figure 1—Overview map of the Late Neoproterozoic to Early Cambrian salt basins of Interior Oman including South Oman Salt Basin (SOSB). Modified after Reuning et al. (2009)

In mature fields of the SOSB, decades of sustained production have resulted in extensive pressure depletion in most producing compartments. Original lithostatic pressure gradients—historically in the range of 18 to 21 kPa/m—have often been reduced to values between 6 and 12 kPa/m in depleted zones. However, certain compartments, particularly those bounded by sealing faults, salt gaps, or heavily plugged matrix, can remain at or near original reservoir pressures despite nearby depletion (Figure. 7).

From a drilling perspective, such lateral compartmentalization poses a high operational risk. Wells planned through zones expected to be severely depleted may suddenly encounter near-original pressures.

This abrupt pressure contrast narrows or eliminates the safe drilling margin between the pore pressure and the fracture gradient, creating the potential for both well control hazards and severe mud losses.

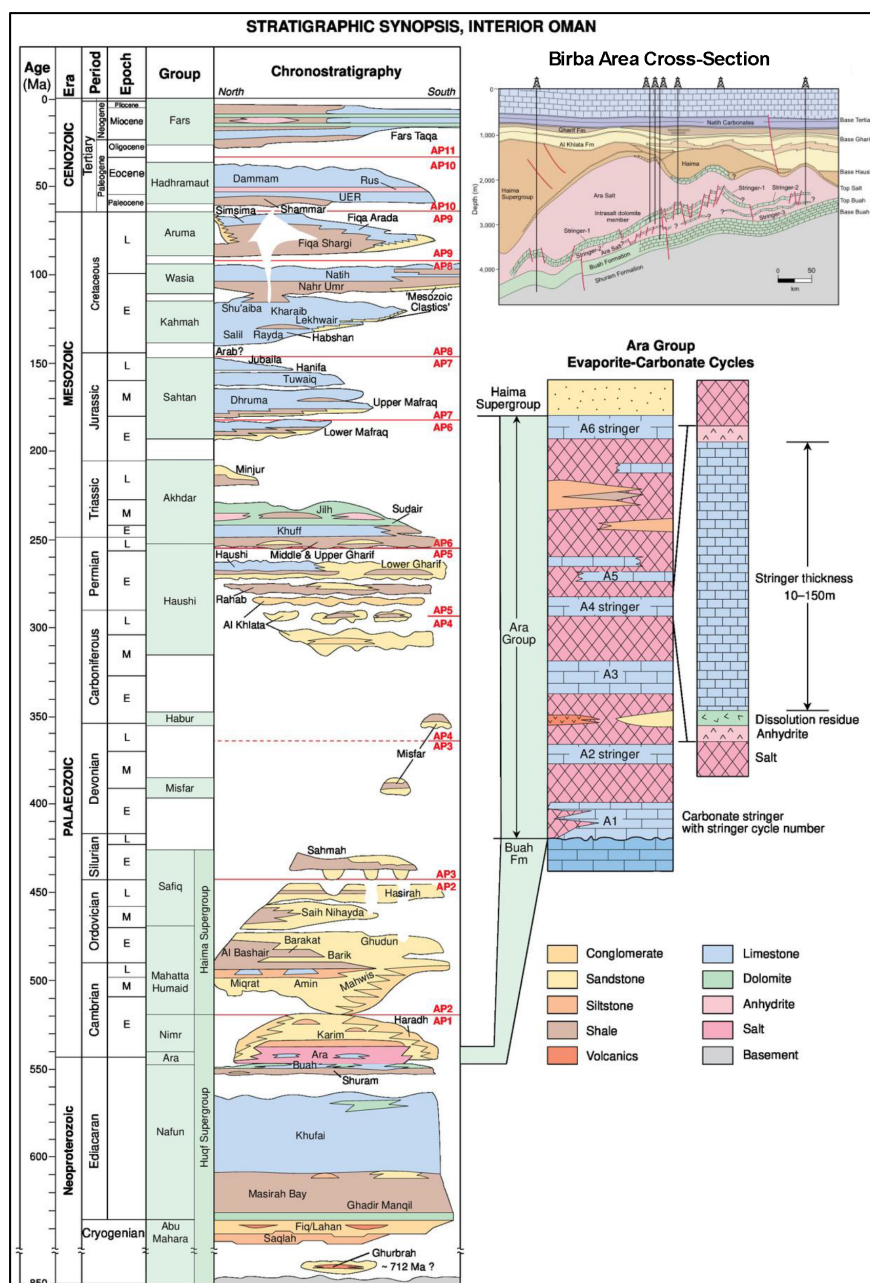


Figure 2—Chronostratigraphic summary of rock units in the subsurface of Interior Oman. The late Neoproterozoic to Early Cambrian is highlighted in the lithostratigraphic column and Birba Area cross-section. Note: single stringer is incised by salt and capped/footed by anhydrite as an evaporitic transition zone. The chronostratigraphic and the lithostratigraphic columns are from Reuning et al. (2009) while the cross-section is from Al-Siyabi (2005).

The WELL-A Incident as a Catalyst

In October 2023, the WELL-A well in the SOSB experienced exactly this scenario. The targeted reservoir block was forecast to have a depleted pore pressure in the range of 8 to 14 kPa/m, based on seismic data, surrounding well data, dynamic models, and the prevailing pore pressure prediction (PPP) model. However, upon penetration, the well encountered an original reservoir pressure of 19.2 kPa/m. The mismatch triggered a kick and required complex control measures.

The investigation into WELL-A identified that the root cause lay not in execution errors but in predictive limitations. The compartment's sealing boundaries had been underestimated in their potential to isolate the block from surrounding depleted zones. The existing PPP workflow did not explicitly account for high-pressure retention in lateral compartments under such geological configurations.

Need for an Integrated Workflow

Prior to WELL-A, the PPP workflow in the SOSB primarily relied on regional pressure trends, well data, dynamic models, and depletion mapping, with qualitative geological interpretation informing the confidence in those predictions. While this approach was effective in many cases, it lacked a standardized mechanism to translate geological boundary characteristics into a quantifiable influence on compartmentalization likelihood.

Key gaps identified included:

- No formal classification system for compartmentalization likelihood.
- Limited integration between geological sealing assessments and PPP modelling.
- Insufficient contingency planning for drilling in areas where the high-pressure case could not be ruled out.

The new workflow addresses these gaps by embedding a qualitative compartmentalization likelihood assessment into the core of the PPP process. This ensures that geological uncertainties are systematically evaluated and directly influence operational planning.

Objectives of the New Approach

Following WELL-A, a dedicated multidisciplinary project was launched with four main objectives:

- Embed learnings from WELL-A into well planning — ensuring that similar scenarios are proactively identified and mitigated.
- Standardize compartmentalization likelihood assessment — creating consistent definitions for Very Unlikely, Unlikely, and Likely cases, based on geological and operational evidence.
- Refine PPP workflows — incorporating high-case original reservoir pressure into drilling designs where warranted by compartmentalization risk.
- Strengthen drilling control and recovery measures — aligning operational procedures with the assessed risk level to ensure drillability, well integrity, and safe exit options.

Scope of This Paper

This paper documents the development, implementation, and validation of the revised workflow in the SOSB Ara reservoirs. The content is organized as follows:

- **Geological and Operational Context** — overview of the SOSB Ara stringers, pressure regimes, and compartmentalization mechanisms.
- **Methodology** — detailed breakdown of the compartmentalization likelihood tool, sealing potential assessment, normalization process, and decision trees for PPP and drillability evaluation.
- **Results** — performance of the workflow in multiple pilot wells.
- **Discussions** — highlights the key strengths of the project and the key limitations
- **Conclusions** — key findings, broader implications for similar reservoirs globally, and recommendations for future applications.

Geological and Operational Context

Regional Framework of the South Oman Salt Basin

The South Oman Salt Basin (SOSB) is a prolific hydrocarbon province characterized by its extensive subsurface salt canopy and the presence of discrete carbonate "stringers" embedded within the Ara Formation salt. The Ara Formation comprises a series of carbonate-evaporite cycles deposited during the late Neoproterozoic to early Cambrian. Each cycle includes a carbonate unit overlain and underlain by anhydrite, forming an evaporitic transition from salt to carbonate and from carbonate back to salt. These anhydrite layers, together with the enclosing halite, bound the stringers both above and below (refer to Figure. 2).

The regional structural framework is dominated by mobile salt tectonics. The ductile salt has migrated in response to differential loading and regional tectonic stresses, creating a variety of traps, closures, and compartmentalized blocks. In plain view, the salt forms a complex network of domes, pillows, and intervening minibasins (Figure-4). The stringers are typically lenticular to tabular in shape and can be isolated or segmented by salt-influenced structural deformation.

Figure-3A illustrates the regional depositional setting prior to stringer deformation, showing their original lateral continuity before salt movement. Figure-3B provides a zoomed-in schematic of the deformation patterns in the Harweel area, highlighting faulted and rotated slabs.

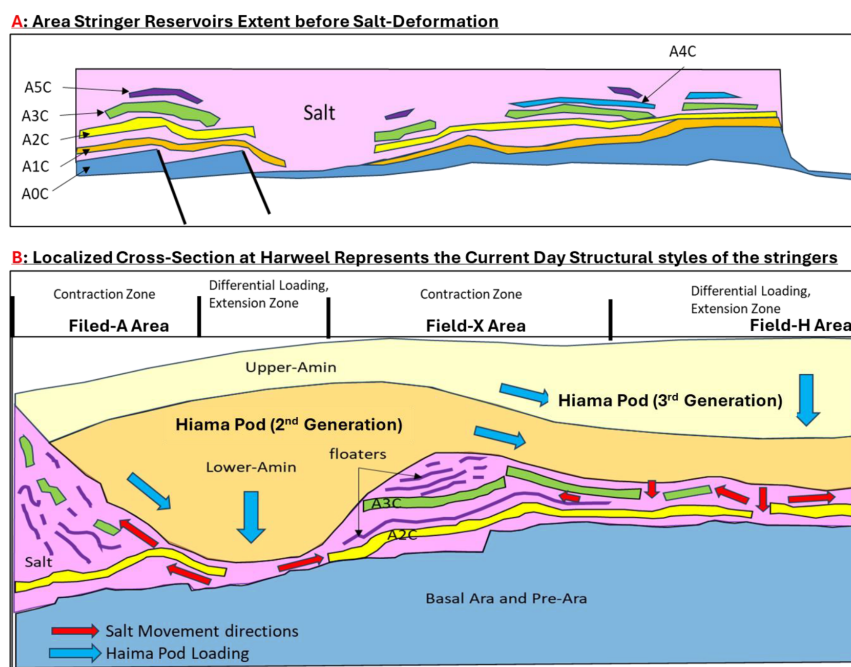


Figure 3—A) illustrates the regional depositional setting prior to stringer deformation, showing their original lateral continuity before salt movement. B) provides a zoomed-in schematic of the deformation patterns in the Harweel area, highlighting faulted and rotated slabs.

Depositional and Structural Evolution of the Ara Stringers

The Ara stringers were deposited in restricted marine settings, producing thick carbonate sequences bounded above and below by anhydrite and ultimately encased in halite. Over time, halokinetic movement altered their geometry, with stringers being rotated, faulted, and locally thinned.

The Haima pods effect — localized pods of denser sediment, dominantly conglomerates — exerted additional lateral and vertical forces during deposition, especially near the top of the salt sequence. This effect intensified with depth due to greater overburden loading, contributing to complex stringer bending and breakage.

The degree of deformation is variable across the basin:

- Deeper stringers (e.g., A2C) tend to be less deformed due to their position closer to the salt base, where mobility is lower.
- Shallower stringers (e.g., A3C and A4C) experience more intense break-up and rotation, increasing the likelihood of compartmentalization.

This observation is supported by field data showing that A1C/A2C reservoirs generally have a lower risk of pressure isolation compared to A3C/A4C reservoirs (see [Figure-4](#): comparison of A2C vs A3C deformation).

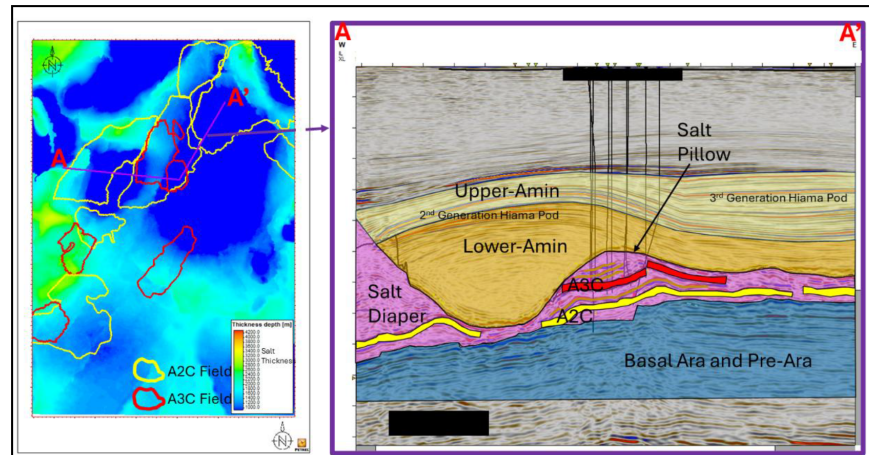


Figure 4—Left) Total Salt Thickness Map overlain by some A3C & A2C Stringers Polygons. The blue color basically represents the Hima Pod plain view geometry while the greenish colors at the side represents Salt diapers and Salt Pillows. Right) A cross-section shows how A3C is deformed in comparison to A2C. The shallower A3C is exposed more to salt-deformation in comparison to the deeper A2C stringer.

Pressure Regimes and Depletion History

Two principal pressure regimes exist within the SOSB Ara stringers:

1. Lithostatic-pressured reservoirs — fully encased in salt, historically holding initial pore pressure gradients of approximately 18–21 kPa/m, as determined from MDT/XPT measurements. These reservoirs are isolated from regional aquifers and retain high pressures unless breached or connected through geological structures.
2. Hydrostatically-pressured reservoirs — in contact or historical communication with underlying aquifers, with typical initial pore pressure gradients of 10–12 kPa/m. Such reservoirs have never held lithostatic pressure or lost it early in geological history through natural discharge.

[Figure. 5](#) Shows the Hydrostatic and Lithostatic reservoir pressure development in Ara reservoir stringers.

With decades of production, pore pressures in most producing compartments have declined into the range of 6–12 kPa/m. However, high-pressure compartments still exist, often associated with:

- Faults connected to salt gaps or field boundaries.
- Zones with high-intensity salt/bitumen plugging.
- Structural highs isolated by sealing faults.

The coexistence of severely depleted and high-pressure compartments within the same reservoir area creates mixed pressure regimes. This presents a major drilling hazard: the safe mud weight window can be

extremely narrow or nonexistent when high-pressure pore cases approach the fracture gradient in depleted sections.

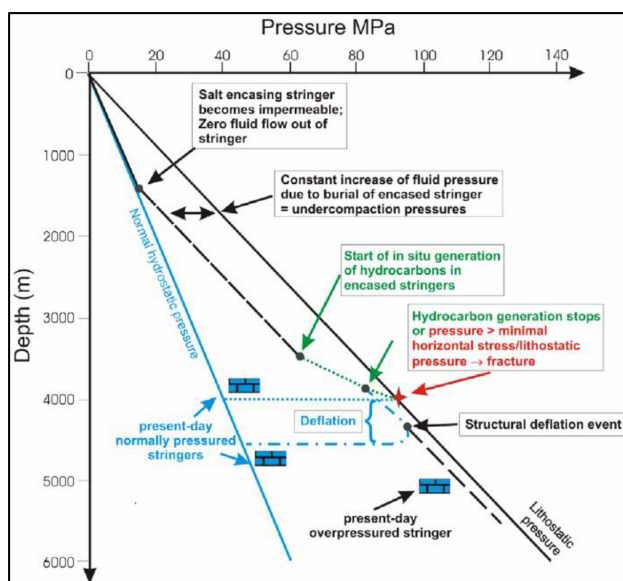


Figure 5—Pressure Development of the present-day Litho-pressured and Hydro-pressured Ara stringer reservoirs. (Khayrutdinov et al. 2023)

Sealing Mechanisms and Compartment Formation

Compartmentalization in the SOSB results from a combination of structural and diagenetic sealing processes:

- **Salt gaps** — Discrete zones where the salt has intruded into or replaced reservoir rock along fault planes, creating highly effective lateral seals.
- **Matrix plugging** — The pore network of the carbonate is occluded by late-stage diagenetic minerals such as halite or bitumen, observed in well logs (e.g., CMR/NMR, sigma logs). High plugging intensity can fully block matrix and fracture permeability.
- **Fault-related sealing** — Faults adjacent to salt boundaries or salt gaps may act as seals due to lateral salt intrusion or creep into the fault plane, or due to diagenetic cementation within fault damage zones.
- **Anhydrite barriers** — The capping and footing anhydrite layers of the stringers (refer to Figure. 2) can play different roles depending on the stress regime (Rowan et al. 2019). In extensional settings, they are more brittle (Rowan et al. 2019) and tend to fracture, allowing the overlying and underlying salt to vertically intrude into the reservoir or its intra-fault zones. In contractional or compressional regimes, the anhydrite behaves more ductile (Rowan et al. 2019), resisting breakage and thereby preventing vertical salt intrusion into the reservoir and its intra-faults.

These mechanisms rarely act in isolation. For example, a single compartment boundary may combine a salt gap with fault-related diagenetic plugging, yielding near-total hydraulic isolation.

Operational Context and Historical Drilling Experience

The Ara reservoirs have been under development for decades, with a drilling history that reflects the dual challenge of depletion and sealing.

Historically:

- Wells drilled through depleted compartments with high mud weights have suffered **severe losses**, particularly when fracture gradients were exceeded in high-permeability intervals.
- Wells encountering unexpected high-pressure compartments have triggered **kicks** requiring well control intervention.
- In certain cases, both high pressure and severe losses were encountered in the same wellbore — a worst-case scenario that can result in loss of the well.

The WELL-A incident stands as a recent and well-documented case where existing PPP methodologies failed to anticipate high-pressure retention in a lateral compartment. This incident, coupled with other earlier but less severe mismatches, reinforced the need for a systematic, basin-specific compartmentalization likelihood assessment as an integral part of PPP.

Implications for Well Planning

Understanding the geological and operational context is essential for well planning in the SOSB. The key implications are:

- The likelihood of compartmentalization varies significantly between reservoir types and structural settings, requiring localized evaluation rather than blanket assumptions.
- Fault-salt interactions and matrix plugging intensity must be explicitly considered in assessing lateral connectivity.
- Mixed pressure regimes demand flexible PPP designs that can accommodate both severely depleted and near-original pressure cases without compromising safety.

These considerations form the foundation for the methodology described in the next section, where geological sealing analysis is systematically integrated into PPP workflows and drillability assessments.

Methodology: Integrated Risk-Based Workflow for Pore Pressure Prediction and Drilling Design

Overview

The revised workflow integrates geological sealing assessment, pore pressure modelling, and drillability evaluation into a single decision framework, allowing drilling strategies to be explicitly linked to compartmentalization likelihood. The process follows three sequential stages:

1. **Stage 1 – Compartmentalization Risk Assessment:** Determine the likelihood that a target block is geologically isolated and retains original reservoir pressure.
2. **Stage 2 – Pore Pressure Prediction (PPP) Range Identification:** Define the pressure envelope, including or excluding original reservoir pressure depending on the assessed likelihood class.
3. **Stage 3 – Drillability Assessment & Operational Pathway Selection:** Compare the required mud weight for safe kick control with available fracture margins, classify as drillable or non-drillable, and apply revision loops where appropriate.

At every stage, the procedures branch according to the classification of the block as **Very Unlikely**, **Unlikely**, or **Likely** to be compartmentalized. This ensures that geological uncertainty is directly reflected in mud weight philosophy, operational readiness, and contingency planning.

The Core Decision-Tree of this project in [Figure. 6](#) summaries the workflow of the three stages.

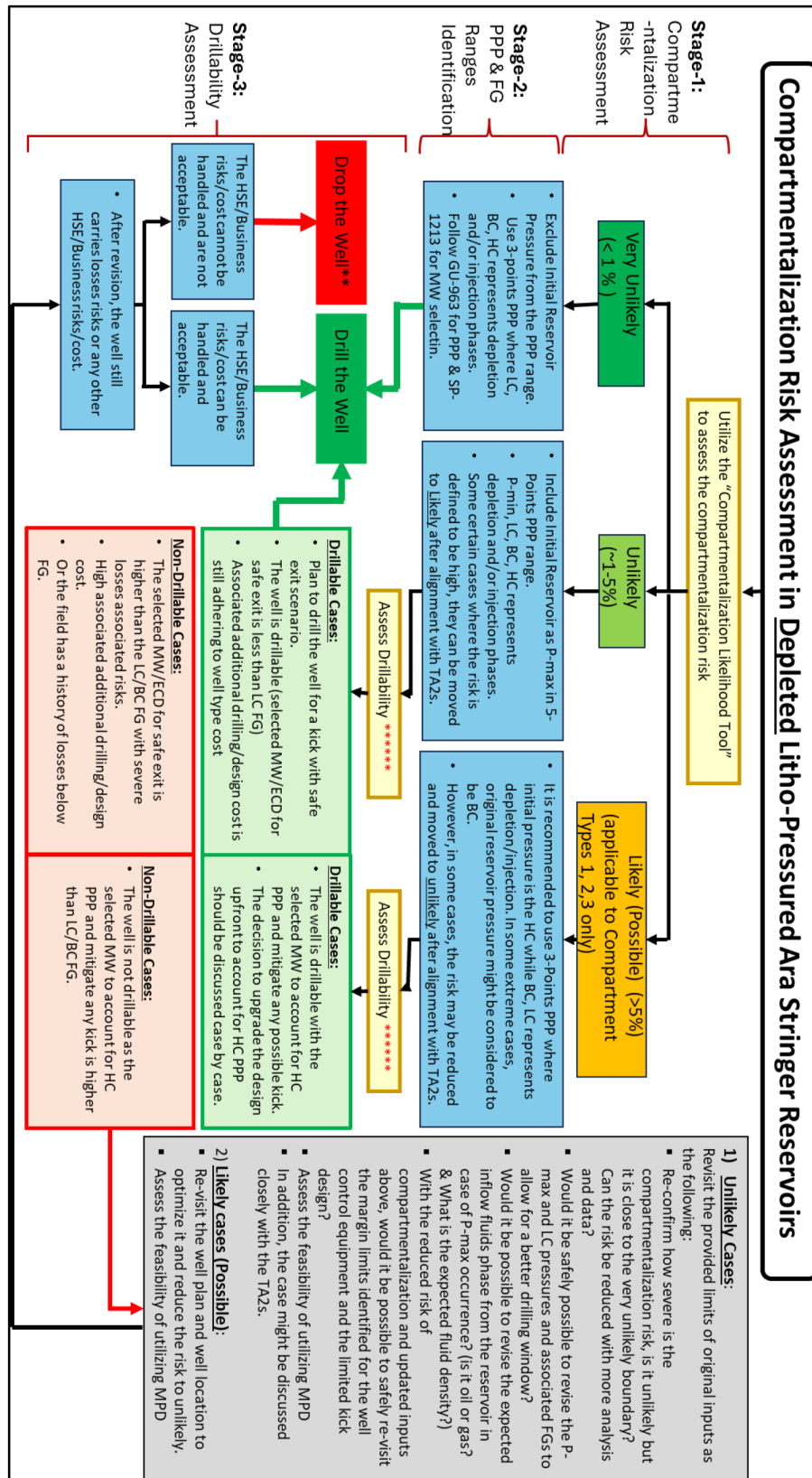


Figure 6—The core Decision-Tree of this project methodology. It summarizes the workflow steps of the three stages (Compartmentalization Likelihood Assessment, PPP, and Drillability Assessment).

Stage 1 – Compartmentalization Risk Assessment

Compartment Types as a Simplification Framework. To streamline the process, the Ara reservoirs are categorized into fourteen compartment types (Figure. 7), each defined by the combination of its boundaries—faults, salt gaps, high-intensity sealed fractures, or fully plugged matrix zones. Each type has a characteristic range of sealing potential derived from historical field data and geological interpretation. This pre-classification enables the tool to start with a baseline likelihood range before incorporating site-specific data.

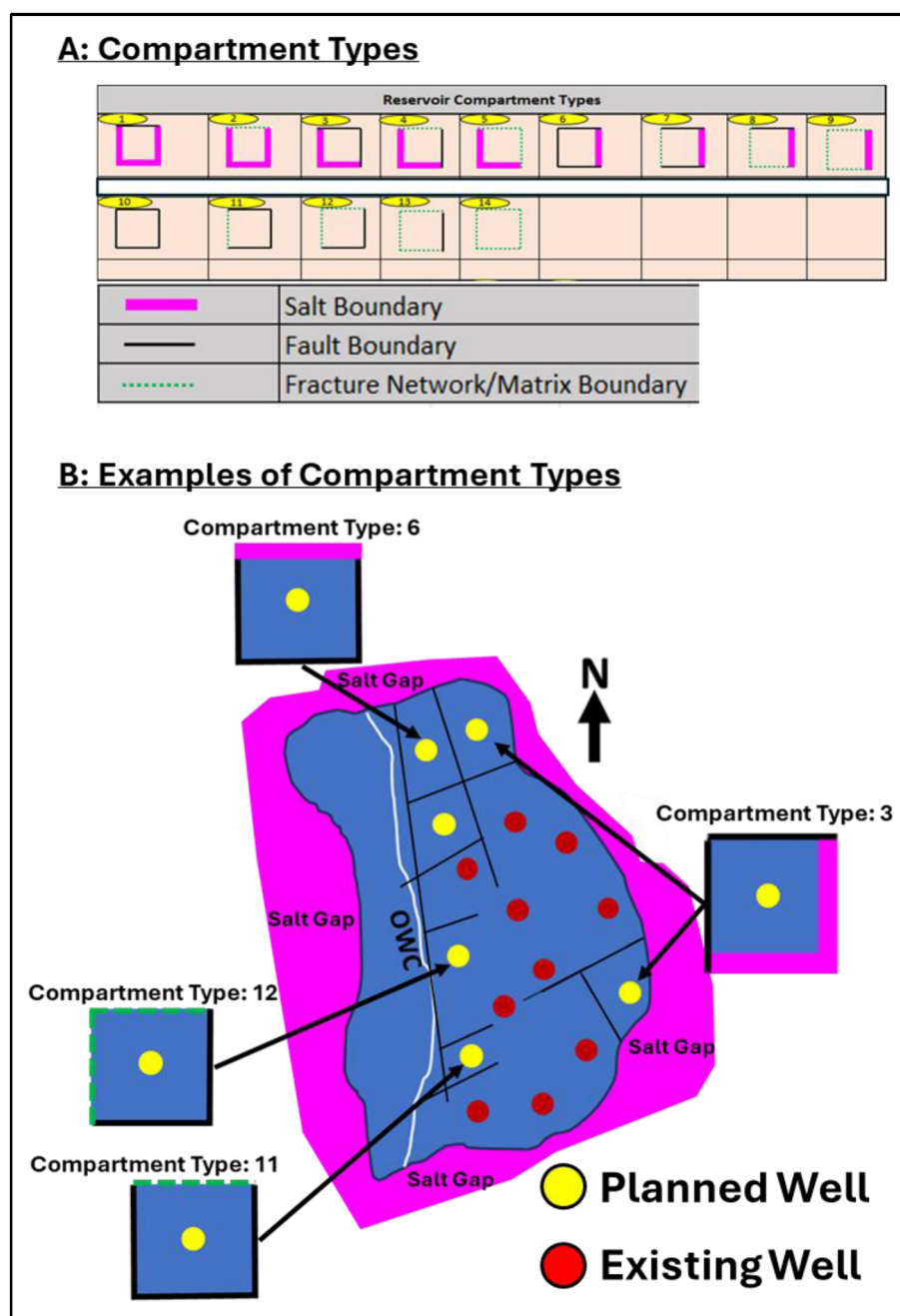


Figure 7—A) A table of the fourteen compartment types which are defined by the combination of the different boundaries. B) A schematic sketch for a stringer field that has existing wells in red, which has depleted the field, and the yellow planned wells that are located within different types of compartments.

Boundary Sealing Potential Assessment. Each boundary of the target block is assigned a Sealing Potential (SP) based on its type:

- **Salt gap boundaries** are treated as near-absolute seals due to the extremely low likelihood of fluid communication through salt.
- **Fault boundaries** are assessed using structured questionnaires, which evaluate indicators such as connection to salt gaps or field boundaries, mapping confidence, displacement magnitude, and associated salt/bitumen plugging intensity.
- **Matrix or fracture boundaries** are evaluated for diagenetic sealing intensity using well logs (e.g., sigma logs for salt plugging, CMR/NMR for bitumen), seismic attributes, and analogue well data.

Compartmentalization Likelihood Formula. The likelihood of a compartment being geologically isolated is calculated using a basin-specific formula:

$$CL = \left(\frac{SP_1 + SP_2 + SP_3 + SP_4}{4} \right) \times NF$$

Where:

- **CL** is the normalized compartmentalization Likelihood
- **SP₁–SP₄** are the sealing potential ratings for each of the four boundaries of the compartment, determined from the assessment in 4.2.2.
- The average SP reflects the overall probability of boundary sealing.
- **NF** is a normalization factor applied to account for geological and operational context.

The tool is designed for four-boundary inputs; non-rectangular compartments are handled by duplicating or combining boundaries to preserve the weighting balance. For example, in triangular compartments with one dominant boundary, that boundary is duplicated in the input set; where boundaries are nearly equal, alternative scenarios are run with different boundaries duplicated.

Normalization Factor. The NF adjusts the un-normalized SP average to reflect conditions specific to the field and reservoir. Key factors include:

- **Reservoir type:** A1C/A2C reservoirs have lower risk than A3C/A4C due to reduced structural deformation.
- **Field history:** Fields with observed compartmentalization—through pressure contrasts, different oil-water contacts, or lack of depletion communication—receive a higher NF.
- **Material balance and dynamic analysis:** Results that confirm or refute isolation can shift the NF up or down.
- **Existing well evidence:** A well within the target block that proves communication or isolation can refine NF.
- **Cross-boundary well data:** Pressure or flow tests from wells across a boundary fault can influence NF if they demonstrate consistent behaviour along the fault trend.
- **Block Size and boundary extension/geometry:** the larger the targeted block with discontinuous boundaries the lower is the compartmentalization likelihood.

Classification Output. The normalized likelihood value from the formula is then compared against defined thresholds to classify the compartment:

- **Very Unlikely:** Geological isolation and high-pressure retention are considered near-impossible.

- **Unlikely:** Geological isolation is possible but of low probability.
- **Likely:** Geological isolation is probable; well planning must assume the potential for original pressure.

Figure. 8 Shows the compartmentalization likelihood matrix that defines the three classifications.

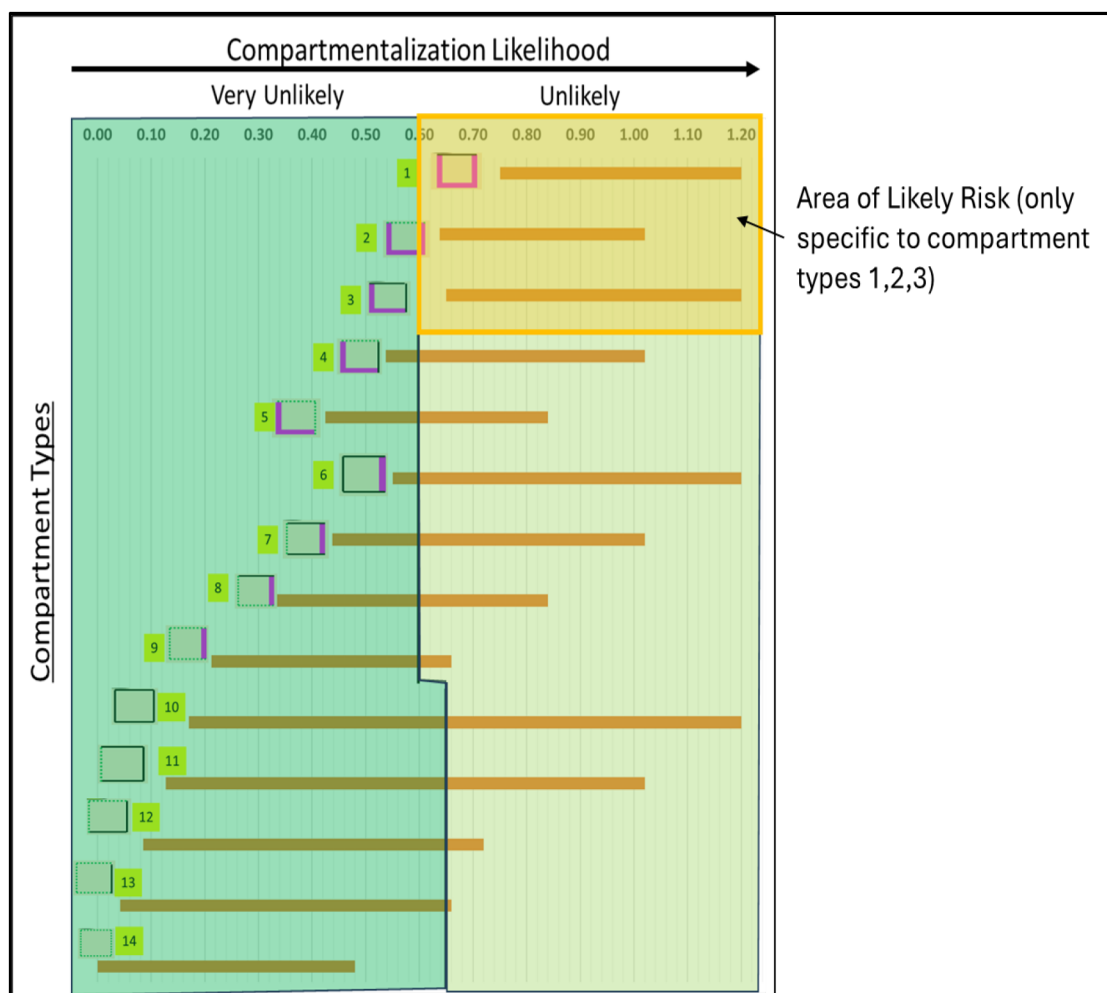


Figure 8—The Compartmentalization Likelihood Matrix which defines the boundary definition between Very Unlikely, Unlikely, and Likely. The brown bars represent the range by which the compartment can fall with after the detailed analysis in the tool. Note: the boundary is different between the compartments that has at least one salt boundary and those that does not have any. These boundaries were defined through intensive sensitivity analysis that is done based on the current and understanding and the historical data of the reservoir compartmentalization in South Oman Salt Basin.

This classification directly informs Stage 2 by determining whether original reservoir pressure (P-max) is excluded, considered as an upper bound, or incorporated as a base/high case in the PPP.

Stage 2 – Pore Pressure Prediction Range Identification

PPP Strategy by Likelihood Class.

- **Very Unlikely:** A 3-point PPP is used, excluding P-max. Low, base, and high cases reflect depletion and/or injection phases only.
- **Unlikely:** A 5-point PPP is used, with P-max as the original reservoir pressure and P-min, LC, BC, HC representing depletion and/or injection phases. P-min is included to represent the most extreme depletion scenario and is particularly useful in fields where miscible gas injection (MGI)

has started, as it enables LC to represent a realistic low-case pressure and improves the drilling window between LC and P-max.

- **Likely:** A 3-point PPP includes P-max as base or high case, assuming high pressure is more probable than depletion.

The PPP outputs from this stage are then passed directly to Stage 3 to evaluate drillability.

Stage 3 – Drillability Assessment and Operational Pathway Selection

Stage 3 converts the outputs from Stages 1 and 2 into a clear operational path by assessing whether the well can be drilled within fracture gradient (FG) limits while safely accommodating the pressure scenarios defined in the PPP range. The decision process is tailored to the Very Unlikely, Unlikely, and Likely classifications, ensuring each receives a fit-for-purpose drilling strategy.

Very Unlikely Cases. In the Very Unlikely category, original reservoir pressure is excluded from the PPP range. MW is optimised for the depletion/injection pressure envelope, prioritising loss avoidance and wellbore stability.

Although P-max is excluded, the rig remains operationally prepared for a surprise influx — well control equipment is on standby, crew readiness is maintained, and kick detection protocols are active. This readiness ensures resilience without unnecessarily elevating MW toward FG limits.

Unlikely Cases. The Unlikely category presents the widest PPP range, spanning from severe depletion (P-min) to original reservoir pressure (P-max). P-min is deliberately included — especially in MGI-affected fields — to ensure that the low-case scenario is realistic and to help position the LC pressure between P-min and P-max for optimal window management.

Key operational elements:

- **Initial MW selection:** Chosen to be high enough to contain a limited-size P-max kick without exceeding LC FG. The minimum MW vs. LC FG chart ([Figure. 9](#)) is used here to illustrate the current status of each field in relation to this criterion, highlighting where LC FG comfortably exceeds the minimum MW and where it approaches or falls below the threshold — the latter representing more challenging P-max management scenarios.
- **Inflow-trigger depth:** This is the first depth where the well is expected to encounter a potential inflow unit. It defines the lower boundary of the P-max readiness interval. If no kick is observed upon reaching this depth, P-max can be excluded.
- **P-max readiness interval:** Drilled at the higher MW from reservoir top to the inflow-trigger depth. The interval length varies — as little as ~10 m in data-rich cases, but significantly longer where logging or formation pressure testing is needed before ruling out P-max.
- **Post-confirmation adjustment:** Once a depleted or injection regime is confirmed, MW is reduced to a level optimized for the narrowed PPP range to mitigate loss risk.

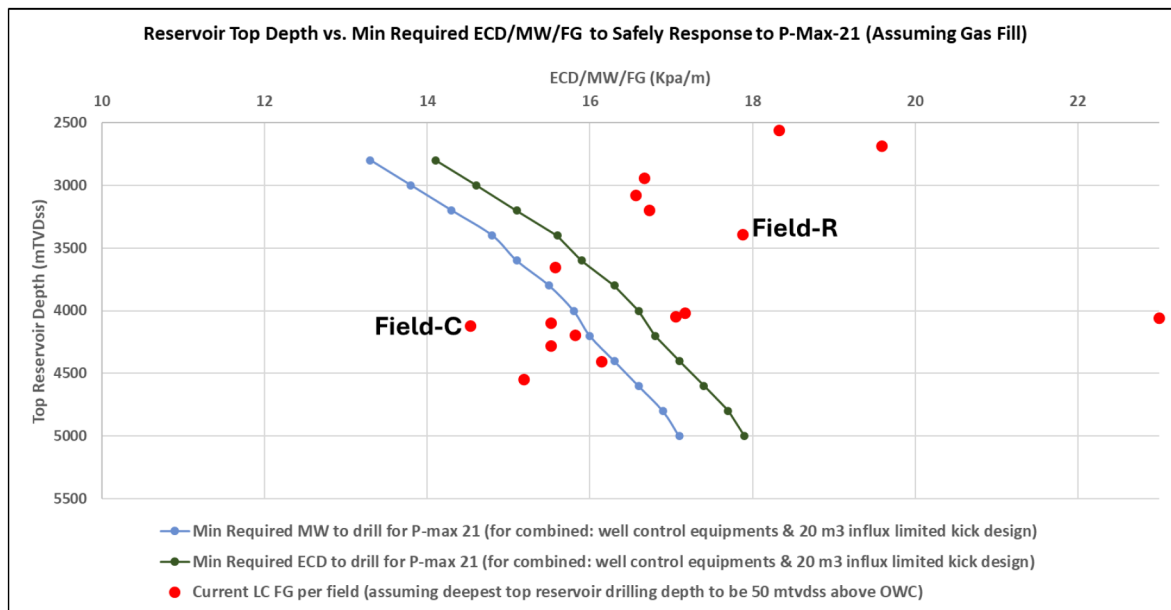


Figure 9—The below graph includes two curves of the minimum required ECD (green) and the minimum required MW (blue) to safely response to P-max of 21 in case it occurred while penetrating depleted litho-pressured reservoir. The red dots show the current LC FG associated with current Low Case PPP of South Oman Salt Basin fields. This shows that the shallow reservoirs such as Field-R has LC FG above the minimum required ECD while the deep and heavily depleted fields such as Field-C falls below the curve.

Revision Loop for Unlikely Cases

If initial screening shows MW requirements above LC FG, the case enters a structured re-evaluation process aimed at creating drillable conditions:

- **Geological re-interpretation:** More granular sealing potential analysis, re-mapping of salt gaps and fault-salt contacts, or targeted generation of advanced seismic attributes to better characterise boundary geometry and sealing behaviour.
- **PPP refinement:** Updating LC and P-max values based on new dynamic model outputs, material balance checks, and direct pressure measurements from offset wells.
- **Inflow-phase analysis:** Determining the expected influx fluid type (oil, gas, or mixed) to adjust kick control MW requirements.
- **Well control capacity review:** Detailed assessment of the well control equipments to determine if lower MW can be safely applied without compromising control integrity.

If revisions bring MW requirements below FG limits, the case is reclassified as drillable; otherwise, target adjustment, well relocation, or deferral is considered. In certain cases, losses/kick risks can still be handled if operationally accepted, then, the well can still be drilled taking these risks.

Likely Cases. In Likely cases, the probability of encountering original reservoir pressure is high enough that the well design fully accommodates P-max from the start. This means:

- MW and casing design accept the loss risk in exchange for kick avoidance.
- Loss control strategies are pre-integrated — such as pre-treatment of loss zones, proactive LCM sweeps, and contingency for cement squeezes.
- Inflow-trigger depth can still be used to roll out original reservoir pressure, but with greater caution before reducing MW, ensuring that all supporting evidence confirms the lower pressure regime before any adjustment is made.

Results

The integrated risk-based workflow was piloted on three wells in the SOSB — WELL-B, WELL-C, and WELL-D — covering two classification categories: *Very Unlikely* and *Unlikely*. While no "Likely" cases were available for testing during the initial implementation period, the available trials provided a robust assessment of how the methodology performs under different risk profiles.

Very Unlikely Case – WELL-B

Classification and Approach. WELL-B was the first well in which the new workflow was applied. The compartment was classified as *Very Unlikely* to be isolated, which under the methodology would normally require a 3-point PPP excluding P-max. However, as this was an early trial, a 5-point PPP including P-max was used, with the *Unlikely* procedures adapted and relaxed to suit the low-risk classification.

Operational Execution.

- P-max readiness interval: Limited to 6 m below top reservoir.
- Initial MW: 14.0 kPa/m — slightly below LC FG (14.5 kPa/m) but sufficient for a short P-max exposure period.
- Pressure parameters: P-max = 21.0 kPa/m; BC = 9.0 kPa/m.
- Post-confirmation adjustment: MW was reduced in two stages to 10–11 kPa/m after P-max was ruled out.

Outcome. The reservoir was under depletion, with no MGI in place, and drilling proceeded smoothly with no incidents or losses. This early application demonstrated that the workflow could be applied flexibly, with reduced readiness intervals in *Very Unlikely* cases, without compromising well control preparedness.

Unlikely Case – WELL-C

Classification and Approach. WELL-C was classified as *Unlikely* to be isolated, requiring inclusion of original reservoir pressure as P-max in the PPP range. The field was already under MGI, making P-min an important input to capture the potential for maximum depletion.

Operational Execution.

- Pressure parameters: P-max = 19.2 kPa/m; BC = 11.0 kPa/m; P-min = 8.0 kPa/m; LC = 9.0 kPa/m.
- FG at P-min: ~16.0 kPa/m.
- Minimum MW for safe P-max drilling: 15.5 kPa/m.
- P-max readiness interval: 15 m below top reservoir.

Once the inflow-trigger depth was reached without evidence of a kick, P-max was ruled out and MW was reduced in two stages to 12.0 kPa/m, optimising the drilling window and reducing loss risk.

Outcome. The well was drilled without incident, validating the workflow's staged reduction approach for *Unlikely* cases and confirming the operational value of incorporating P-min in MGI fields.

Unlikely Case with Revision Loop – WELL-D

Classification and Initial Challenge. WELL-D (drilled as gas injector) within Field-C (Refer to [Figure 9](#)) represented a severe depletion scenario, with an extreme spread between LC PPP (5.6 kPa/m) and P-max (21.0 kPa/m). The initial assessment showed that the MW required for safe P-max containment exceeded LC FG, making the case borderline drillable.

Revision Loop Application. The well underwent the full revision loop process from Stage 3:

- **Geological refinement:** Seismic attribute analysis reduced the assessed risk of heavy salt plugging.
- **Material balance analysis:** Showed compartmentalization was impossible under base STOIP and only possible under high-case STOIP.
- **Risk adjustment:** Classification remained Unlikely but shifted toward the Very Unlikely boundary.
- **Influx phase analysis:** Expected kick fluid identified as oil, reducing MW requirement for kick control.
- **MW adjustment:** Minimum MW reduced from 16.0 to 14.0 kPa/m, just below LC FG (14.2 kPa/m).
- **Inflow-trigger depth:** Reduced from 30 m to 10–15 m below top reservoir based on seismic attribute analysis correlated with existing well logs.
- **Loss mitigation readiness:** LCM and cement plugs prepared on site.

Figure 10 shows how the risk of compartmentalization reduced in the block of WELL-D after detailed analysis through the "Revision Loop".

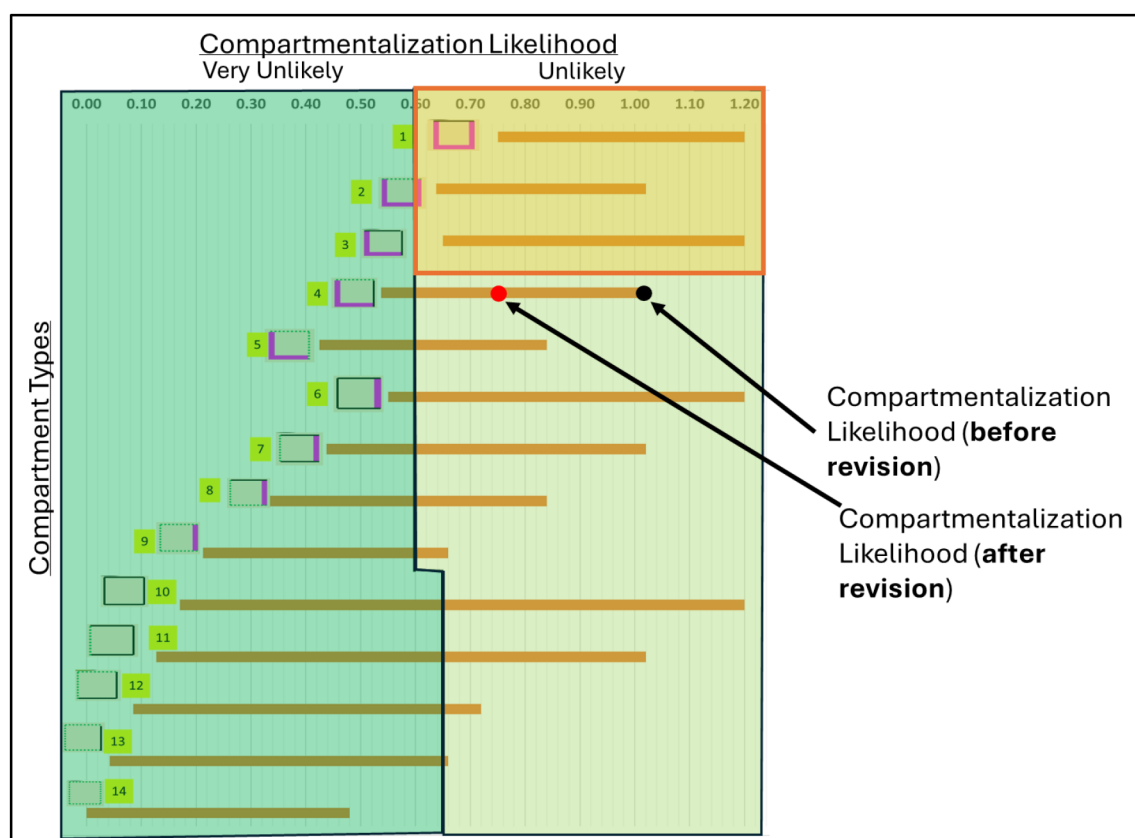


Figure 10—The compartmentalization likelihood of WELL-D was reduced after going through detailed analysis by applying the "Revision Loop".

Outcome. The well was drilled smoothly without incidents or losses. Compared to historical analogues in similar fields, where overbalanced drilling caused severe losses and extended NPT, WELL-D demonstrated a substantial operational improvement.

Overall Workflow Impact

Across the trial wells:

- No P-max surprises were encountered.
- MW reduction strategies after ruling out P-max were consistently implemented without compromising safety.
- Fields previously associated with high loss risk during overbalanced drilling showed improved outcomes, with zero severe losses in these cases.
- The workflow's flexibility allowed tailoring of readiness intervals and MW strategies to the assessed likelihood class, while the revision loop converted borderline non-drillable cases into successful operations.

Discussions and Limitations

Strengths of the Methodology

The implementation of the Compartmentalization Likelihood Framework and risk-based PPP workflow represents a major step forward for PDO's well delivery system in the South Oman Salt Basin. Key strengths include:

- **Standardization of Geological Uncertainty:** Moves decision-making from subjective expert judgment to structured, repeatable scoring logic.
- **Alignment of Geology with Drilling Risk:** Integrates sealing potential and structural complexity directly into PPP and MW design.
- **Operational Flexibility:** Decision trees allow dynamic MW adjustments and safe exit strategies without compromising HSE integrity.
- **Cost Efficiency:** Optimizes well design by avoiding unnecessary overdesign while mitigating catastrophic failure risk.

Limitations

Despite its benefits, the methodology is not without constraints:

- **Data Dependency:** High reliance on accurate fault interpretation and salt-gap detection. Seismic quality and imaging uncertainty can skew SP scoring.
- **Subjectivity in SP Rating:** While scoring matrices reduce bias, assigning confidence levels to diagenetic indicators (e.g., plugging intensity) can remain subjective.
- **Sparse Calibration Points in Frontier Areas:** In under-explored blocks, lack of historical compartmentalization data can lead to conservative NF values, inflating perceived risk.
- **Dynamic Behavior Post-Depletion:** Stress-path coefficient (α) and FG response vary spatially, requiring continuous LOT calibration to refine FG prediction models.

Conclusions

The development and field application of a compartmentalization risk-based planning and drilling workflow in South Oman Salt Basin have delivered clear improvements in safety, operational efficiency, and predictive accuracy. This outcome stems from the structured integration of geological risk assessment into engineering design, coupled with adaptive operational execution.

Key conclusions are as follows

1. Geological Understanding is Foundational

- The Ara stringer reservoirs of the South Oman Salt Basin present a unique combination of lithological heterogeneity, salt tectonic deformation, and diagenetic sealing.

- Pressure isolation arises from multiple mechanisms — salt gaps, fault–salt interactions, matrix plugging, and fracture occlusion — which can act independently or in combination.
- Not all boundary types carry equal sealing potential; systematic evaluation using predefined criteria (SP ratings) is essential to avoid over- or underestimating isolation risk.
- Reservoir type (A1C, A2C, A3C, A4C) and compartment geometry (e.g., size, boundary continuity) influence the likelihood of compartmentalization and must be factored into risk normalization.

2. Risk-Based PPP Envelopes Improve Preparedness

- Moving from deterministic, depletion-trend-based PPP models to risk-adapted envelopes ensures that the full spectrum of plausible pressures is considered in well design.
- The three- or five-point PPP models, chosen according to CL classification, capture both high-side (P_{max}) and low-side (depletion) scenarios without inflating uncertainty unnecessarily.
- This approach reduces the likelihood of encountering pressures outside the design envelope and aligns drilling readiness with actual geological risk.

3. Safe-Exit Mud-Weight Design Expands Drillable Opportunities

- The safe-exit strategy provides a practical compromise between fracture avoidance and well-control readiness in narrow-margin wells.
- Designing MW to allow well suspension and upgrade if P_{max} is encountered avoids pre-emptive overdesign while retaining safety margins.
- Defining inflow trigger depths allows timely exclusion of P_{max} , enabling MW reduction to minimize loss risk without sacrificing preparedness.

4. Structured Decision Trees Enable Consistent, Rapid Response

- Embedding clear decision pathways for each CL category reduces the reliance on ad-hoc judgement under pressure.
- In-field execution benefits from predefined escalation points, such as MW increase, liner setting, or suspension, based on real-time indicators.
- The ability to reclassify compartmentalization risks of a compartment through detailed revision loop, as demonstrated in WELL-D, prevents premature abandonment of drillable opportunities.

5. Measurable Operational Gains

Application of the methodology in the pilot wells produced tangible improvements compared to historical offsets:

- No well-control incidents or severe losses.
- PPP predictions closely aligned with measured pressures.
- Drilling efficiency improved through early optimization of MW profiles.

These gains were achieved without compromising process safety or reservoir integrity, underscoring the value of integrating subsurface risk assessment with operational planning.

6. Transferability and Wider Application

- The workflow is designed for adaptation to other salt-tectonic carbonate basins, provided it is calibrated to local geological conditions.
- SP criteria and NF values should be redefined using local fault-salt geometries, diagenetic signatures, and compartmentalization histories.
- The safe-exit MW philosophy, PPP range logic, and decision-tree framework are inherently transferable and can be embedded into existing drilling governance processes.

- o Early integration of multidisciplinary expertise — geoscience, drilling, well integrity, and reservoir engineering — is essential for successful adoption.

Final Statement

The success of this approach in SOSB demonstrates that the uncertainty posed by compartmentalization in salt-encased carbonate reservoirs can be managed proactively. By embedding geological risk into every stage of well planning and execution, operators can safely expand the range of drillable opportunities, reduce operational surprises, and improve overall field development efficiency. With careful calibration, this methodology offers a robust template for managing similar challenges in geologically complex basins worldwide.

Nomenclature

BC	Base Case pore pressure, kPa/m
CL	Compartmentalization Likelihood
CMR	Combinable Magnetic Resonance log
FG	Fracture Gradient, kPa/m
HC	High Case pore pressure, kPa/m
LC	Low Case pore pressure, kPa/m
LCM	Lost Circulation Material
LOT	Leak-Off Test
MDT/XPT	Modular/Extended Formation Dynamics Tester
MGI	Miscible Gas Injection
MW	Mud Weight, kPa/m
NF	Normalization Factor
NMR	Nuclear Magnetic Resonance log
NPT	Non-Productive Time
P-max	Maximum pore pressure (original reservoir pressure), kPa/m
P-min	Minimum pore pressure (maximum depletion case), kPa/m
PPP	Pore Pressure Prediction
SOSB	South Oman Salt Basin
SP	Sealing Potential rating
STOIIP	Stock Tank Oil Initially In Place
TVDSS	True Vertical Depth Subsea

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